Summary in Graph

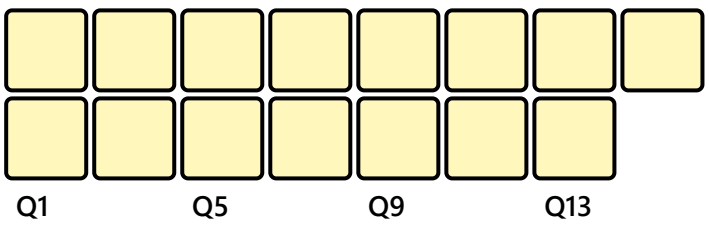
Exam Summary (GO Classes CS/DA | Linear Algebra | Weekly Quiz 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (9 + 6)
Total Marks:	21 (9 + 12)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (42) out of 419 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Linear Algebra

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a 2×2 real matrix. Which of the following is a necessary and sufficient condition for A not to have a real eigenvalue?

A. $(a-d)^2 + bc < 0$

B. $(a+d)^2 - 4(ad-bc) < 0$

C. $ad - bc < 0$

D. $a^2 + d^2 < 4bc$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Find the values of $\alpha \in \mathbb{R}$ such that the matrix

$$\begin{bmatrix} 3 & \alpha \\ \alpha & 5 \end{bmatrix}$$

has 2 as an eigenvalue.

A. $\alpha = \pm 1$

B. $\alpha = \pm\sqrt{2}$

C. $\alpha = \pm\sqrt{3}$

D. $\alpha = \pm 2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $A = (a_{ij})$ be a real 3×3 matrix such that

$$a_{ij} = -a_{ji} \quad \text{for all } 1 \leq i, j \leq 3$$

Suppose $3i$ is an eigenvalue of A. What are the other eigenvalues of A ?

A. $-3i$ and 0

B. $3i$ and $-3i$

C. $-3i$ and 3

D. 3 and -3

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $x \in \mathbb{R}^n$ be a non-zero column vector, and define

$$A = xx^T$$

What is the necessary and sufficient condition for the matrix

$$Q = I - 2A$$

to be an orthogonal matrix?

A. x is an eigenvector of A
B. x is a unit vector (i.e., $\|x\| = 1$)
C. x has all entries equal to 1
D. $\|x\| = 0$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $\mathbf{x}$ and $\mathbf{y}$ be two non-zero $n \times 1$ column vectors. Let $A = \mathbf{x}\mathbf{y}^T$ be an $n \times n$ matrix. What are the eigenvalues of A ?

A. All eigenvalues are zero.

B. Exactly one eigenvalue is $\mathbf{y}^T\mathbf{x}$, and the rest are zero.

C. All eigenvalues are equal to $\mathbf{y}^T\mathbf{x}$.

D. The eigenvalues are $\|\mathbf{x}\|\|\mathbf{y}\|$ and $n-1$ zeros.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let M be a real symmetric 3×3 matrix that admits the vectors

$$\mathbf{v}_1 = (1,2,3)^T \quad \text{and} \quad \mathbf{v}_2 = (1,1,-1)^T$$

as eigenvectors.

Which of the following vectors is surely an eigenvector of M ?

A. $(1,-1,0)^T$

B. $(-5,1,1)^T$

C. $(3,2,1)^T$

D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 3×3 matrix with eigenvalues $-1, 1, 2$. Suppose there exist real scalars α, β, γ such that:

$$A^{-1} = \alpha A^2 + \beta A + \gamma I$$

Then find the value of:

$$\alpha + \beta + \gamma = ?$$

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a real 3×3 upper triangular matrix with diagonal entries $1, 2, -3$.
Suppose A^{-1} can be written as a linear combination of I, A, A^2, i.e.,

$$A^{-1} = \alpha I + \beta A + \gamma A^2$$

Find the value of $12(\alpha + \beta + \gamma)$.

Your Answer: | Correct Answer: 12 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{m \times n}$ be a real matrix of rank m, where $n > m$. Suppose that for some real number α, the following holds:

$$AA^\top = \alpha I_m$$

Then, which of the following statements about
$A^\top A \in \mathbb{R}^{n \times n}$ are true?

A. $A^\top A$ has exactly zero distinct eigenvalues

B. $A^\top A$ has eigenvalue 0 with multiplicity$n - m$

C. α is an eigenvalue of $A^\top A$ with multiplicity at least $n - 2$

D. $A^\top A$ has exactly two distinct eigenvalues

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements are true?

A. There exists a 2×2 real orthogonal matrix that has 2 as an eigenvalue.

B. There exists a 2×2 real orthogonal matrix that has i as an eigenvalue.

C. If $A \in \mathbb{R}^{2 \times 2}$ is orthogonal, then $\|Ax\| = \|x\|$ for every $x \in \mathbb{R}^2$, where $\| \cdot \|$ denotes the length of a vector.

D. If $A \in \mathbb{R}^{2 \times 2}$ is orthogonal, then all its eigenvalues are real.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given that the matrix

$$A = \begin{pmatrix} \alpha & 1 \\ 2 & 3 \end{pmatrix}$$

has 1 as one of its eigenvalues, what are the trace and determinant of the matrix?

A. Trace $= 4$, Determinant $= 5$

B. Trace $= 5$, Determinant $= 4$

C. Trace $= 6$, Determinant $= 3$

D. Trace $= 3$, Determinant $= 6$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Pick out the true statements from the following:

A. For any two $n \times n$ matrices A and B,

$$(A+B)^2 = A^2 + 2AB + B^2$$

B. There exist $n \times n$ matrices A and B such that

$$AB - BA = I$$

C. For any invertible $n \times n$ matrix B, and any matrix A.

$$\operatorname{tr}(A) = \operatorname{tr}\left(B^{-1}AB\right)$$

D. If A is a real symmetric matrix, then all its eigenvalues are real.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 03sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be a 4×4 real matrix with eigenvalues $-1, 1, 1, -2$. Define

$$B = A^4 - 5A^2 + 5I$$

Then the value of $\mathrm{tr}(A+B)$ is:

A. 0

B. -12

C. 3

D. 9

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $n \times n$ real matrix such that the set of all its non-zero eigenvalues has exactly r elements. Which of the following statements is true?

A. $\mathrm{rank}(A) \leq r$

B. If $r = 0$, then $\mathrm{rank}(A) < n - 1$

C. $\mathrm{rank}(A) \geq r$

D. A^2 has r non-zero eigenvalues

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A and B be $n \times n$ real matrices, and let

$$C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}$$

Which of the following statements are true?

A. If λ is an eigenvalue of $A+B$, then λ is an eigenvalue of C.
B. If λ is an eigenvalue of $A-B$, then λ is an eigenvalue of C.
C. If λ is an eigenvalue of A or B, then λ is an eigenvalue of C.
D. All eigenvalues of C are real.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

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